

Semiconductor Memory

1. Compare NAND and NOR flash wrt the following features.
 - Storage density
 - Random/Sequential Read
 - Random/Sequential Program

NAND typically faster read/write, blocks are bigger than NOR
2. The SSD controller manage the translation between Logical Block Address (LBA) from the Application processor to the Physical Block Address (PBA) of the actual storage location of the data in the SSD.
 - Table 2.1 shows the mapping of LBA to PBA.
 - For simplicity's sake, we can assume that these addresses at block level address, e.g. LBA block 0 (LBA0) is mapped to PBA block 9 (PBA9) in the table.
 - Table 2.2 shows the erase count of each PBA block.
 - LBA, PBA block correspond to the flash block in the flash structure, i.e. erasure is at block level.
 - A 'Y' in the 'Empty' column indicates that the Flash Block is empty.
- (a) Complete Table 2.1 for Dynamic wear leveling algorithm. You can further assume that
 - Access sequence from the processor is LBA 7, 4, 3, A.
 - All access will go through the wear levelling algorithm for physical block assignment.
 - If the LBA was originally assigned to a different PBA, the PBA containing the old data will be erased.

Step by step what happens:
Eg. LBA7 is currently in PBA7- look for Empty == Y and lowest erase count (PBA2)
LBA7 =>2, PBA2 Empty = N, PBA7 erase => 0x79, N=>Y

What is the consideration to take for the erasure process of the old data?

Table 2.1	
LBA	PBA
0	9
1	X
2	X
3	5
4	X
5	4
6	6
7	7
8	X
9	X
A	0

Table 2.2		
Erase Count Table		
PBA	Erase Count	Empty
0	5	N
1	33	Y
2	15	Y
3	98	Y
4	45	N
5	11	N
6	3	N
7	78	N
8	A3	Y
9	F1	N
A	56	Y

- (b) In table 2.2, you can see that PBA block 6 has a very low erase count but would not be evaluated under Dynamic wear levelling algorithm. Static wear levelling attempts to relocate such flash blocks that are seldom re-written to achieve more even erase count distribution. There are many ways to implement Static wear levelling, below is one example. Go through the algorithm described and suggest enhancement to the algorithm.

- On every access, all blocks (occupied and empty) will be evaluated for block assignment.
- PBA block with the lowest erase count will be selected to store the data.

Can configure max no. of hops, after a certain number, it will just look for empty blocks only

3. Complete the timing diagrams in Figure 4.1 and 4.2 below, fill in the missing information on commands, data, addresses etc.

- Other than supplying address information, what are other functions of A10 signal?
- What is the CAS latency of the SDRAM for Figure 4.1?

In SDRAM, u only can access different rows in same bank 1 by 1, row1 need to close before accessing row 2 in bank 1. But different banks has no issues, bank 1 and bank 2 can be read same time if databus allows.

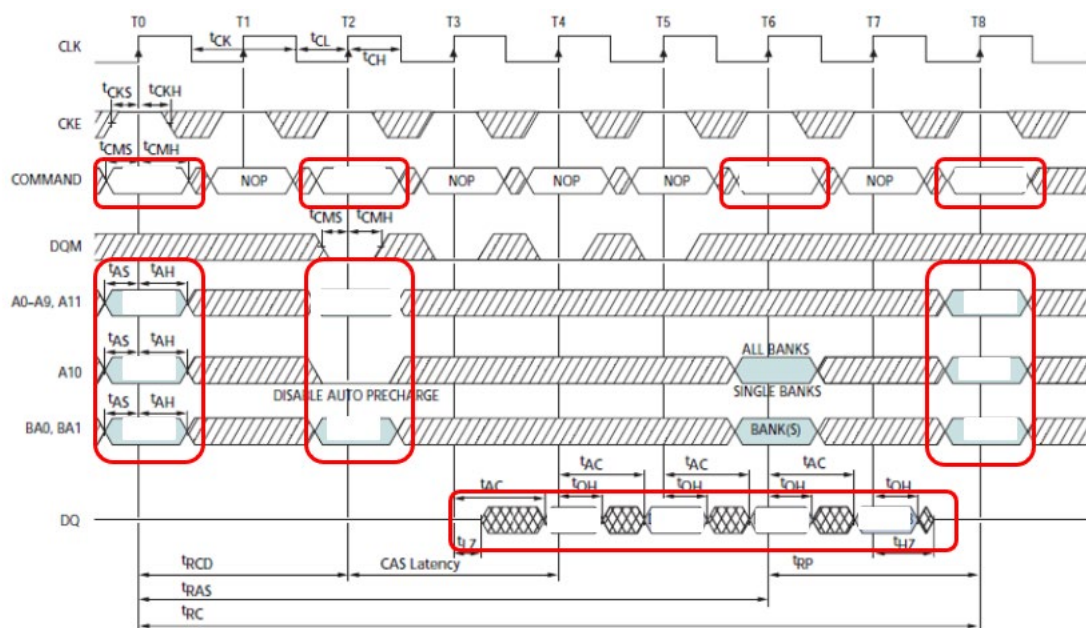


Figure 4.1: SDRAM Read without Auto-Precharge

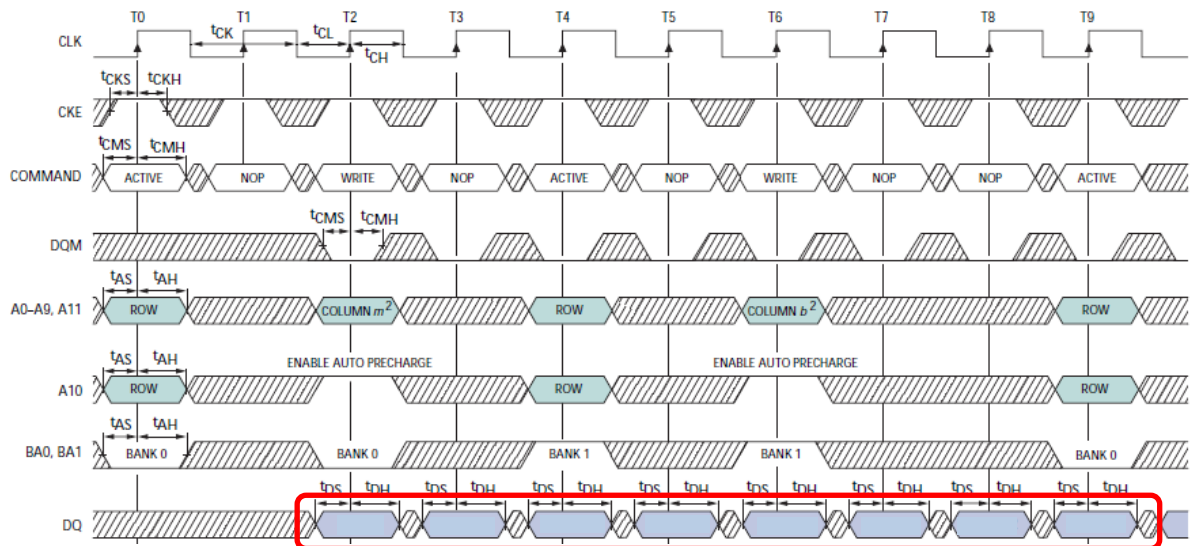


Figure 4.2: SDRAM Write – Alternating Bank Access (with Auto-Precharge)

4. What is the difference between auto-refresh and self-refresh operation? Reference Figure 5.1 below, t_{REF} parameter for this SDRAM, which has 8192 rows, is 64ms max. What would be the expected t_{REF} value for a SDRAM with 4096 rows? Assuming same SDRAM design and technology is used.

Parameter	Symbol	-7E		-75		Unit	Notes
		Min	Max	Min	Max		
Refresh period (8192 rows)	t_{REF}	–	64	–	64	ms	
AUTO REFRESH period	t_{RFC}	66	–	66	–	ns	

Figure 5.1: SDRAM Auto-Refresh Operation Parameters

(Optional)

5. Figure 3.1 and 3.2 shows the read and write access timing diagram of a SRAM. It is the signal that SRAM expects or output during a Read/Write operation. When a processor interface to the SRAM, it needs to output signals conforming to what the SRAM expect and needs to read in signals SRAM generates, all within specified timings e.g. t_{OE} , t_{OHZ} etc. Describe the significance of the following parameters during Read/Write operation, and whether the parameters should be a min or max value.
- t_{OE} , t_{OHZ}
 - t_{WP} , t_{DH}

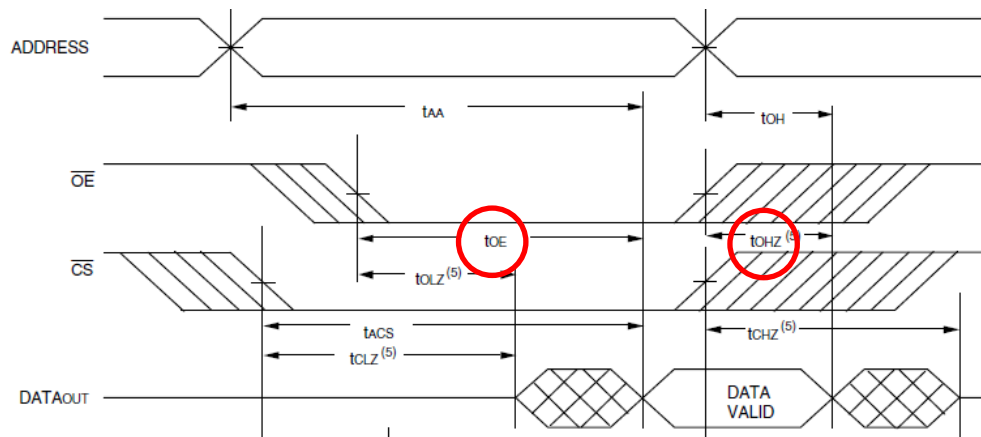


Figure 3.1: SRAM Read access timing diagram

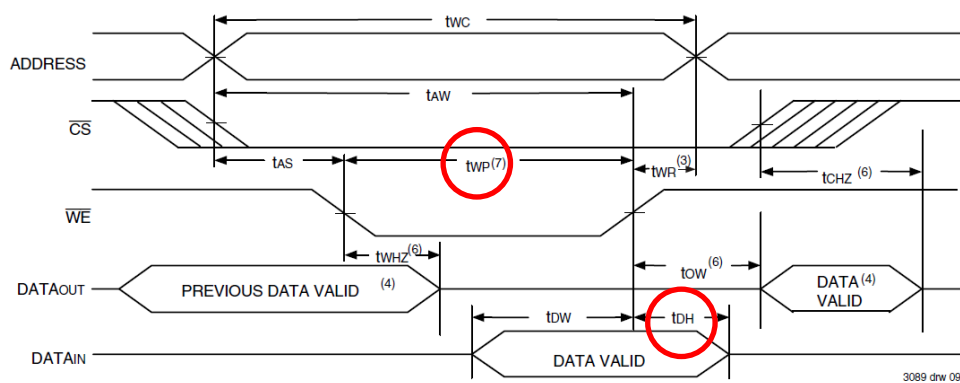


Figure 3.2: SRAM Write access timing diagram

6. SSD Design considerations: homework given during lecture.